GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

Interior and Exterior Angles of a Triangle

Independent Set 6.T2 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Two angles of a triangle are 63° and 79°. Find the third angle.

38°

180 − 142.

2. An isosceles triangle's two base angles are equal. The angle between its two equal sides is 40°. Find each base angle.

70°

(180 − 40) ÷ 2.

3. A triangle's remote interior angles are 46° and 58°. Find the exterior angle.

104°

46 + 58.

4. An exterior angle of a triangle is 140°. One remote interior angle is 62°. Find the other.

78°

140 − 62.

5. A triangle's angles are $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$. Find each angle.

40°, 60° and 80°

9x = 180, x = 20.

6. An exterior angle of a triangle measures $(4x+10)^\circ$; its remote interior angles measure 50° and $(2x)^\circ$. Write an equation and find x.

4x + 10 = 50 + 2x; x = 20

2x = 40.